Kirill Kozlov

Junior AI Engineer (LLM, RAG, OCR, Python)

Moscow, Russia (GMT+3), open to remote work and relocation
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Professional Summary

Software engineer and Machine Learning master's student at CMC MSU with production experience at X5 Tech. Built Python services using FastAPI, PostgreSQL, and Kubernetes; developing AI product search and catalog-processing pipelines for Korzina.

Technical Skills

LLM and RAG: Large Language Models (LLM), prompt engineering, ML pipelines, LLM agents, model selection and configuration, RAG (Retrieval-Augmented Generation), BM25, LangChain, OpenAI API, Ollama, Claude Code
Documents and OCR: OCR (Optical Character Recognition), vision-language models for scanned documents, key entity extraction and attribute validation from business documents, PyMuPDF, python-docx
Experimentation: A/B testing (X5 Tech summer school, 2026)
Programming Languages: Python, SQL, Go, C++
Backend: FastAPI, SQLAlchemy (async), Alembic, Pydantic, pytest, asyncio, httpx, aiohttp
Data: PostgreSQL, Redis, Apache Kafka
Delivery and CI/CD: Docker, Kubernetes (K8s), Helm, GitLab CI/CD, Grafana, Git, Linux
Languages: Russian — Native, English — B2

Work Experience

Software Engineer (Junior) | X5 Tech (X5 Group, Russia's largest grocery retailer), Logistics Solutions Department, Moscow | October 2026 — Present

- Developing the system that handles 90% of all foreign supplies X5 Retail Group purchases for its retail chains.

Machine Learning Developer (Intern) | X5 Tech, AI Agents Department, Moscow | April 2026 — October 2026

- Built and deployed a multi-agent LangChain contract-review pipeline with parallel document analysis: 500+ production reviews in May–August 2026; legal opinions in 8–19 minutes versus approximately 97 minutes of manual work; 8 FTE impact.
- Selected and configured LLMs for the pipeline: commercial (OpenAI GPT-4o-mini) and open-source models served via Ollama in a private corporate environment (Llama 3.2, Qwen 2.5, Qwen3-VL for scanned documents).
- Set up a RAG-oriented data source over Russian legislation: 40 federal laws, 366 articles (including the Civil Code), BM25 index, relevant articles injected into agent prompts.
- Extracted key entities from contracts, property registry extracts, and powers of attorney using OCR and LLMs; validated consistency across documents.
- Built and deployed to dev and prod an OCR and LLM service that reads foreign invoices for X5 Tech: document conversion, multi-pass OCR, LLM-based field extraction; in pilot since August 2026.
- Developed the FastAPI backend for utility-bill processing: OCR results consumed from Apache Kafka, stored and served to the frontend; 10 FTE saved.
- Own delivery end to end: technical design, FastAPI services, Docker images, Helm charts, GitLab CI/CD deployment to Kubernetes (dev and prod), post-deployment latency monitoring in Grafana.

Projects

Korzina — Marketplace and AI Product Search | TypeScript, Node.js, PostgreSQL, React | September 2026 — Present

- Building a Telegram Mini App and bot with a catalog of 100,000+ products; automated supplier-data ingestion, normalization, and deduplication.
- Implemented PostgreSQL full-text and trigram search; developing a hybrid retrieval prototype combining multimodal Gemini embeddings, pgvector, reciprocal rank fusion (RRF), and Qwen reranking.
- Implemented LLM/VLM attribute extraction and product-compatibility checks in the prototype, with versioned embedding caches, concurrent-request deduplication, and incremental refresh.
- Built offline relevance evaluation with frozen datasets, qrels, Recall@20, and Precision@5; compared text and multimodal retrieval under model-cost and latency constraints.
- Built the Fastify API and background jobs; deployed the service using PostgreSQL, Valkey, S3, Docker, and GitLab CI/CD.

Other projects: AxiomAI: LLM-powered Telegram bot, 80% reduction in manager workload; WB Parser: Wildberries parser and Mini App, 220 users.

Education

Lomonosov Moscow State University, Faculty of Computational Mathematics and Cybernetics
Master's degree in Machine Learning, Dept. of Mathematical Methods of Forecasting | September 2026 — Present
Bachelor's degree | August 2022 — August 2026
Additional training: X5 Tech Summer School on A/B Testing, 2026 (certificate); Deep Learning School, MIPT (Computer Vision and NLP).